

TrES-3b

R-2011

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130421 · created 2026-08-07T17:11:44+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456403.8136 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.118 +/- 0.042
Transit depth (Rp/Rs)^2	1.39 +/- 0.99 [%]
Orbital Inclination (inc)	82.7 +/- 1.48
Airmass coefficient 1 (a1)	1.206 +/- 0.016
Airmass coefficient 2 (a2)	-0.145 +/- 0.0049
Scatter in the residuals of the lightcurve fit is	1.28 %
Best Comparison Star	None
Optimal Aperture	3.43
Optimal Annulus	14.1
Transit Duration (day)	0.054 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.118 $\pm$ 0.042 vs 0.16309 (deviation 1.07 σ)
Duration fit vs published	0.054 d vs 0.0567 d (deviation 0.17 σ)
Mid-time O-C	-3.4 min
Depth SNR	2.8
Residual scatter	1.28 %

Light curve

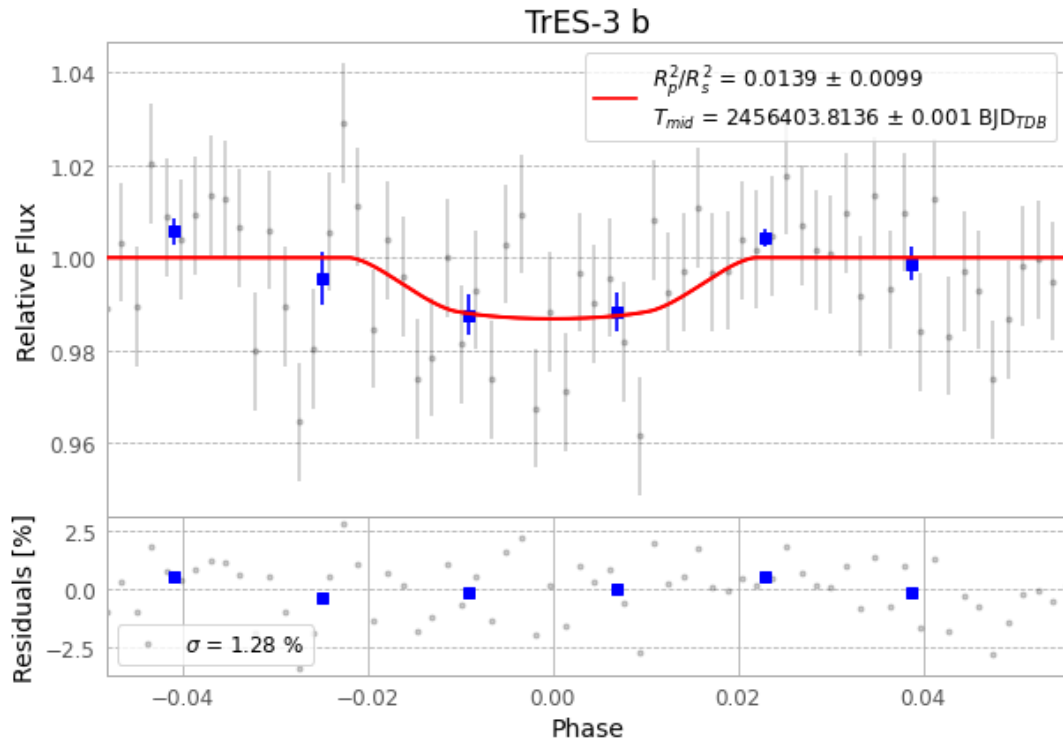


Figure 1 — FinalLightCurve_TrES-3 b_2013-04-20.png (SHA-256 5c14f39abae287b3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [244, 173], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 883c70973aa988b9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

66 FITS frames (43 MB), TRES-3130421055712.FITS → TRES-3130421091213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130421.log (8f2685ccdc05...), FinalLightCurve_TrES-3 b_2013-04-20.png (5c14f39abae2...), FinalLightCurve_TrES-3 b_2013-04-20.csv (2f8a11710342...)

Compute resources

Wall time 118.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 17:11Z.